

Let $\frac{1-x}{x}$ be $y-1$. $\therefore \frac{1}{x}-1=y-1 \Rightarrow x=\frac{1}{y}$ $x-1=\frac{1}{y}-1=\frac{1-y}{y}$

$3f\left(\frac{1-y}{y}\right)-f(y-1)=\frac{2}{y} \Rightarrow \boxed{3f\left(\frac{1-x}{x}\right)-f(x-1)=\frac{2}{x}} \quad (2)$

$\boxed{3 \times \text{eqn 1} + \text{eqn 2}}$: $8f(x-1) - \cancel{3f\left(\frac{1-x}{x}\right)} + \cancel{3f\left(\frac{1-x}{x}\right)} - f(x-1) = 6x + \frac{2}{x}$

$8f(x-1) = 6x + \frac{2}{x} \Rightarrow \boxed{f(x-1) = \frac{3x}{4} + \frac{1}{4x}}$. Let $x-1=y \Rightarrow x=y+1$

$f(y) = \frac{3(y+1)}{4} + \frac{1}{4(y+1)} \Rightarrow \boxed{f(x) = \frac{3}{4}(x+1) + \frac{1}{4(x+1)}}$

Prob 9: Find analytical relations and the natural domains of the following functions:

a) $f(1-x)$; b) $f\left(\frac{1}{x}\right)$ for $f(x) = \log(x^2-1)$

$f(x) = \log(x^2-1)$ a) $f(1-x) = \log((1-x)^2-1) = \log(x^2-2x)$ $x^2-2x > 0 \Rightarrow x(x-2) > 0$
 $x > 0$ and $x-2 > 0 \Rightarrow x > 2$
 $x < 0$ and $x-2 < 0 \Rightarrow x < 2$

Domain: $[-\infty, 0) \cup [2, \infty[$

b) $f\left(\frac{1}{x}\right) = \log\left(\frac{1}{x^2}-1\right) = \log\left(\frac{1-x^2}{x^2}\right) = \boxed{\log(1-x^2) - \log(x^2)}$

$1-x^2 > 0$ $x^2 > 0$
 $|x| < 1$ $x \neq 0$ $[-1, 1] \rightarrow \text{Domain}$